

SYNTHESIS PROJECT

An AprilTag-Based Approach to Trajectory Recording and Relocalization for Mobile Robots

Joao Pedro Martins do Lago Reis and Yann Kelvem da Silva Ramos

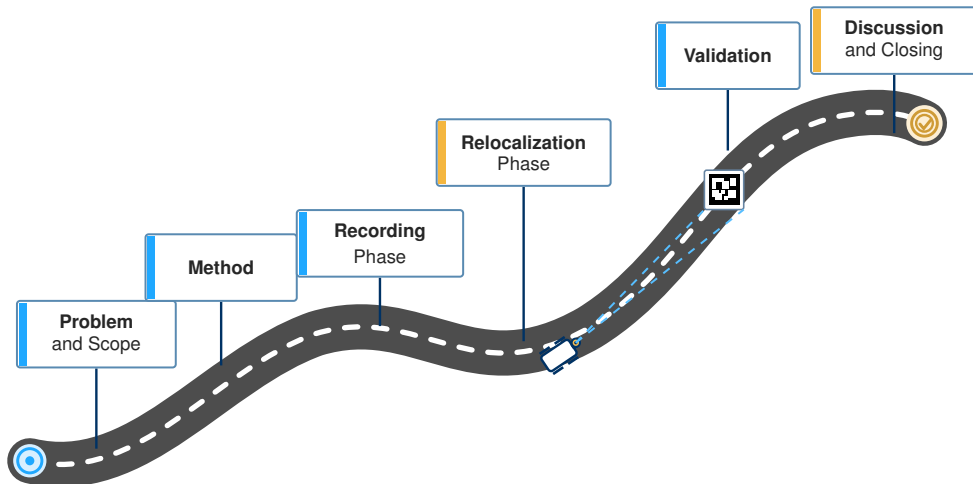
Master's degree in artificial perception and robotics

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Supervisor: Dr. Sébastien Lengagne

Evaluator: Dr. Aufrère Romuald

Outline



Problem and Scope

For whom

Polytech practical-work students using AgileX LIMO in indoor AprilTag setups.

Why

Need a reproducible workflow to record a path and recover pose relative to it for control-law exercises.

Problem

Recording alone is insufficient because odometry drift breaks trajectory consistency across runs.

Scope

In: offline reconstruction and relocalization outputs (D , ϕ).
Not delivered here: trajectory tracking controller implementation.

Problem and Scope

Scope: control law design is future work

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To fill later:

- ▶ Objective of the method
- ▶ Pipeline blocks
- ▶ Main equations
- ▶ Expected outputs

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Recording Phase

Recording Phase — Template

To fill later:

- ▶ Inputs and sensors
- ▶ Offline optimization steps
- ▶ Generated files
- ▶ Key interpretation points

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Relocalization Phase

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- ▶ Nearest-path logic
- ▶ Distance/heading outputs
- ▶ RViz visualization items

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- ▶ Simulation scenario
- ▶ Quantitative metrics
- ▶ Comparison criteria

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Discussion and Conclusion — Template

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- ▶ Main findings
- ▶ Current limitations
- ▶ Scope boundaries
- ▶ Next steps

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Conclusion

Thank you for listening!

Any questions?

Joao Pedro Martins do Lago Reis | Yann Kelvem da Silva Ramos
Record, Map, and Relocalize with AprilTags